PHARMACEUTICAL ANALYSIS

UNIT 5 NOTES

ELECTROCHEMICAL ANALYSIS

- CONDUCTOMETRY
- POTENTIOMETRY
- POLAROGRAPHY

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CONDUCTOMETRIC TITRATION

Conductometric Titration is an electrochemical method of analysis used for the determination of concentration of unknown solution on the basis of its electrical conductivity by means of a **Conductometer.**

PRINCIPLE

- During the titration process, the conductivity of the solution changes due to the replacement or movement of Ions.
- These changes are monitored and plotted against the volume of titrant added.
- The end point is identified as the point where a significant change in conductivity occurs.

SOME IMPORTANT DEFINITIONS

① **Conductance :** Conductance is the ability of a material or solution to conduct electric current. It is reciprocal of resistance and is denoted by symbol G.

$$G = 1/R$$

② **Resistance :** Resistance is the property of a material that opposes the flow of electric current through it. It is denoted by R

$$R \propto l/a \quad \text{or} \quad R = \rho l/a$$

- Here ρ is specific Resistance

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③ **Specific Conductance** : It is defined as conductance of a solution measured per unit length & cross sectional area. Also known as 'kappa'.

$$K = \frac{l}{a} G$$

FACTORS AFFECTING CONDUCTIVITY

- Types of Ions
- Concentration of Ions
- Temperature
- Mobility of Ions

MOLAR CONDUCTIVITY OF SOME IONS

- H^+ : 349.6
- OH^- : 199.1
- Cl^- : 76.35
- NH_4^+ : 73.5
- NO_3^- : 71.46
- Ag^+ : 61.9
- Na^+ : 50.1
- CH_3COO^- : 40.9

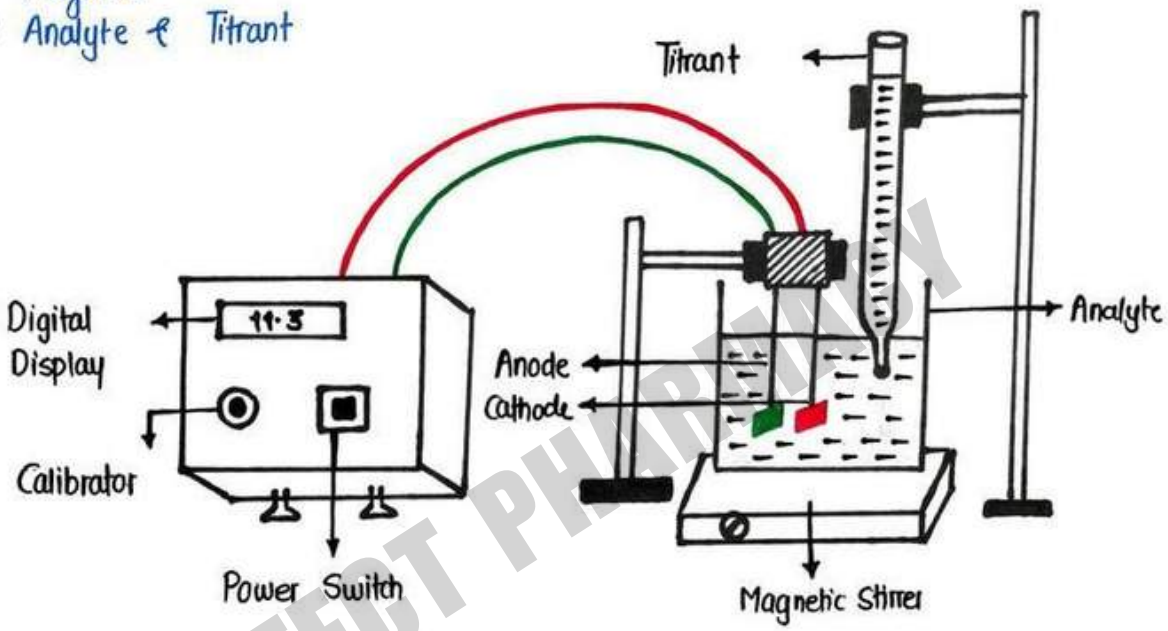
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INSTRUMENTATION OF CONDUCTOMETRY

It consist of a :

- Conductometer
- Conductivity Cell
- Magnetic Stirrer
- Analyte & Titrant



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① CONDUCTOMETER

It consist of a

- Digital Display
- Calibrator
- Power Switch

- Alternative Current (AC) is used as source

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② Conductivity Cell

- It is made of Pyrex or Quartz
- It is fitted with two electrodes
  - ⓐ Cathode
  - ⓑ Anode

③ Electrodes

- Electrodes are made of Platinum
- Area = 1 cm^2
- Distance b/w Electrodes = 1 cm

Procedure

- Prepare the analyte solution & immerse conductivity electrodes.
- Measure the initial conductivity.
- Add titrant in small increments, stirring the solution after each addition.
- Record the conductivity after each addition.
- Plot a graph of conductivity v/s titrant volume
- Identify the equivalence point from the graph.

Application Of Conductometry

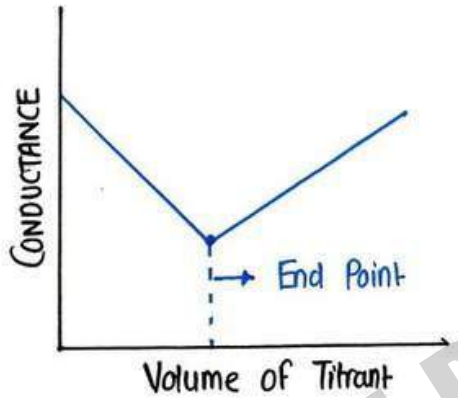
- Check Water Pollution in rivers & lakes
- Determination of Alkalinity of Fresh water
- Determination of Mixture of Acids.

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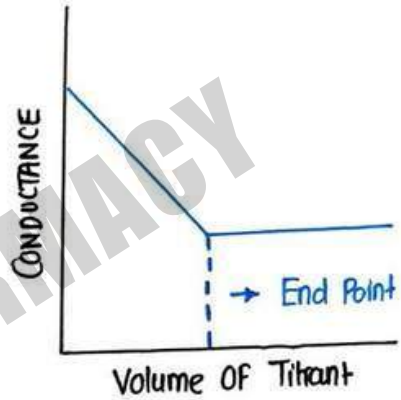
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Types Of Conductometric Titration

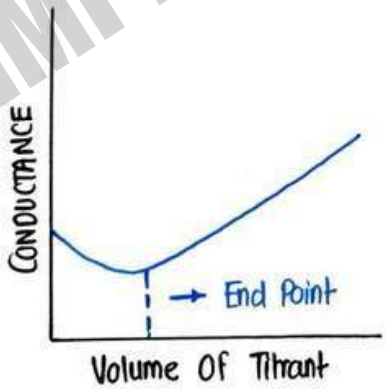
- Strong Acid vs Strong Base
- Strong Acid vs Weak Base
- Weak Acid vs Strong Base
- Weak Acid vs Weak Base



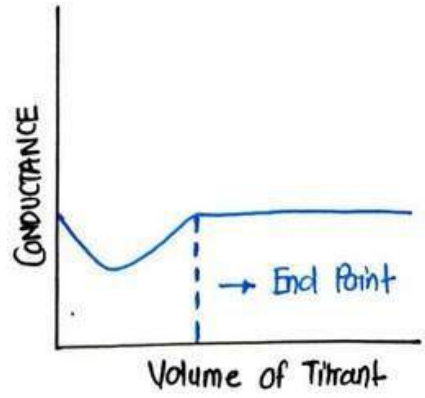
① Strong Acid V/s Strong Base



② Strong Acid V/s Weak Base



Weak Acid V/s Strong Base



Weak Acid V/s Weak Base

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POTENTIOMETRY

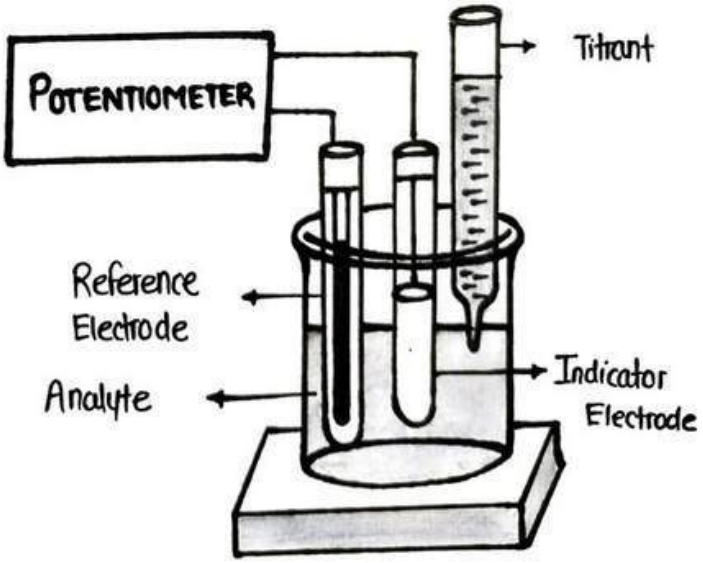
Potentiometry or Potentiometric Titration is an analytical technique used to determine the concentration of an analyte in solution by measuring the change in the electric potential (voltage) of a solution during titration process.

PRINCIPLE

- Potentiometric Titration works by measuring the change in voltage (electric potential) between two electrodes as a chemical reaction takes place during the titration.
- One Electrode (Reference Electrode) keeps a constant voltage, while the other (Indicator Electrode) respond to changes in the concentration of ions in the solution.

INSTRUMENTATION

- Indicator Electrode
- Reference Electrode
- Analyte
- Titrant
- Potentiometer



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PROCEDURE

- Prepare the analyte solution and place it into the titration vessel
- Insert the reference and indicator electrodes into the solution.
- Start adding the titrant slowly while stirring.
- Record the potential changes at regular intervals or continuously
- Plot a graph of electrode potential v/s volume of titrant added.
- The end point of the titration is the point at which analyte has completely reacted with the titrant, It is indicated by a sharp change in the potential difference on the titration curve.

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REFERENCE ELECTRODES

- A Reference Electrode is an essential component of potentiometric titration that provides a stable and constant potential against which the potential of the indicator electrode can be measured.
- It does not participate in the chemical reaction but serves as a fixed point for accurate voltage comparison throughout the titration process

TYPES OF REFERENCE ELECTRODE

① Standard Hydrogen Electrode
② Saturated Calomel Electrode
③ Silver Chloride Electrode.

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① STANDARD HYDROGEN ELECTRODE

It is a reference electrode used in potentiometry to measure the potential change during the titration reaction.

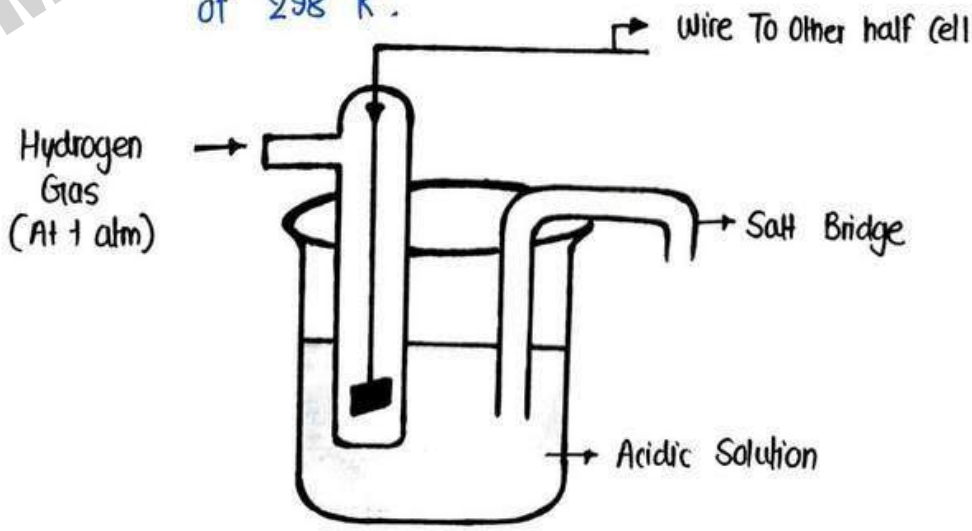
CONSTRUCTION

① Platinum Electrode : • A platinum wire or sheet coated with platinum black is used as electrode.
• Platinum is chemically inert and conducts electron efficiently.

② Hydrogen Gas : Pure hydrogen gas is bubbled over the platinum electrode at a constant pressure of 1 atm.

③ Acidic Solution : The electrode is immersed in an acidic solution, typically containing 1 M hydrogen ions (H^+) from a strong acid like HCl.

④ Temperature : The electrode operates at a standard temperature of 298 K.



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APPLICATION

- It is used as primary reference for measuring the electrode potential during titration.
- It is essential in the construction of electrochemical cells.

Advantages

- Provides a stable reference potential
- Can be easily calibrated.

Disadvantages

- Handling is risky due to flammability of Hydrogen Gas
- Platinum can be expensive & contaminate.

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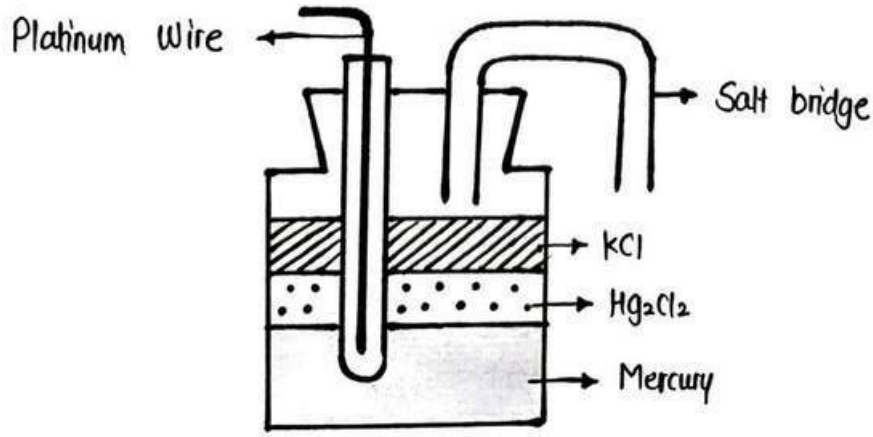
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② SATURATED CALOMEL ELECTRODE

- The saturated calomel electrode is widely used reference electrode in potentiometric titration due to its stable and well defined potential .
- It provides a reliable reference point for measuring the electrode potential of various half cells.

CONSTRUCTION

- Glass Container : A glass vessel holds the components and includes a side arm for contact with external solution.
- Mercury : Pure liquid mercury is placed at the bottom of the container , serving as electrode material.
- Calomel Paste : A layer of mercury chloride (Hg_2Cl_2) is added above mercury.
- Saturated kCl Solution : The container is filled with a saturated potassium chloride solution to maintain constant chloride ion concentration.



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APPLICATION

- Electrochemical Studies
- Reference electrode in potentiometry
- pH Measurment
- Corrosion Testing

ADVANTAGES

- Stable Potential
- Easy Maintainance

Limitations

- Mercuric Toxicity
- Temperature Sensitivity

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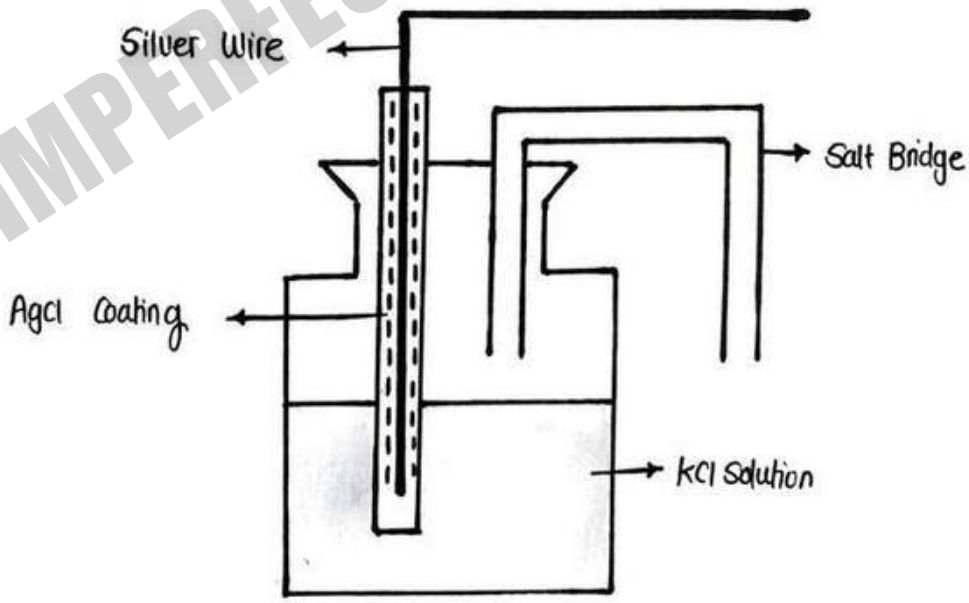
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③ SILVER CHLORIDE ELECTRODE

- The Silver - Silver Chloride electrode is a widely used reference electrode used in electrochemical measurements due to its stable and well defined potential.
- It is known for being a safer and more environmental friendly alternative to the saturated Calomel Electrode.

CONSTRUCTION

- Electrode Body: Usually made of glass to hold the components.
- Silver Wire: A pure silver wire is coated with a thin layer of silver chloride through chemical deposition.
- KCl Solution: The electrode is immersed in a fixed concentration of KCl solution.



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APPLICATION

- pH Measurment
- Potentiometric Titration
- Corrosion Monitoring

ADVANTAGES

- Stable Potential
- Environmentally safer
- Temperature Compatability

DISADVANTAGES

- Electrolyte Leakage
- Chloride Ion Dependence

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POLAROGRAPHY

Polarography is an electrochemical analytical technique that measures the current flowing through a solution as a function of an applied voltage, typically using a dropping mercury electrode as the working electrode.

PRINCIPLE

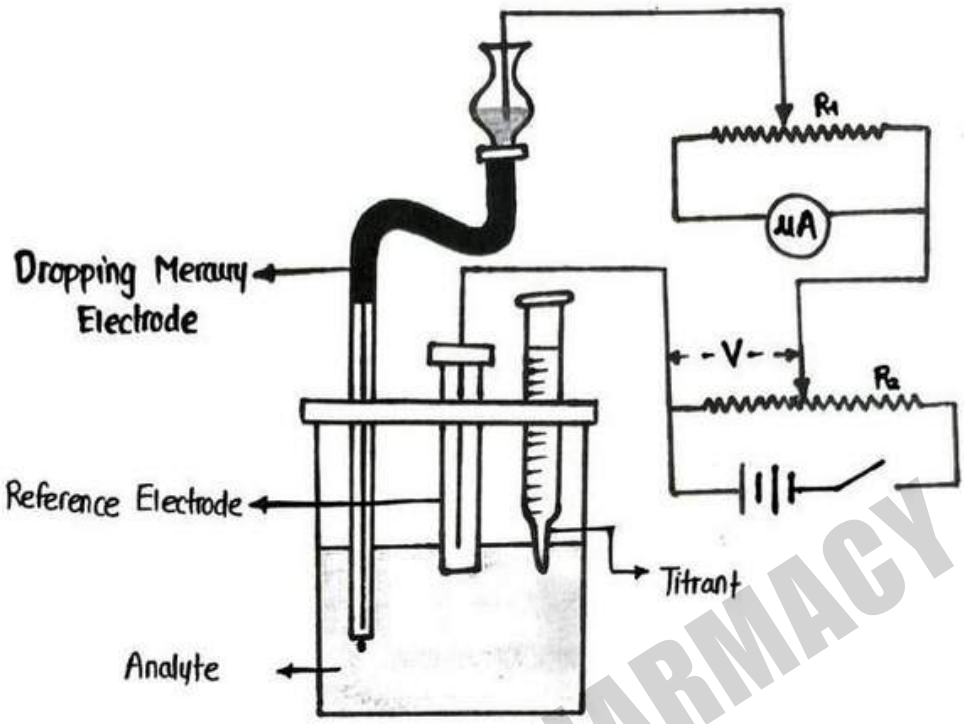
- In polarography a linearly increasing voltage is applied between the working electrode and a reference electrode.
- When the applied potential reaches the point where a chemical in the solution starts to gain or lose electrons, a noticeable current change occurs.

COMPONENTS

- Dropping Mercury Electrode (DME): Consist of a glass capillary through which mercury flows, froming droplets at its tip.
- Reference Electrode: Commonly a saturated calomel electrode.
- Analyte: Solution to be analysed.
- Polarograph: Instrument that applies voltage and measures current.

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PROCEDURE

- Prepare the analyte solution.
- Setup the polarographic cell with the electrodes properly immersed.
- Apply a linearly varying voltage across working & reference electrode
- Record the current.
- Measure the concentration using Ilkovic equation

$$i_d = 607 \times n \times D^{1/2} \times m^{2/3} \times t^{1/6} \times C$$

- Here i_d = Diffusion current
  - n = No of electrons involved in redox reaction
  - D = Diffusion Coefficient
  - m = Rate of mercury drop from the electrode
  - C = Concentration of analyte
  - t = Drop time of mercury

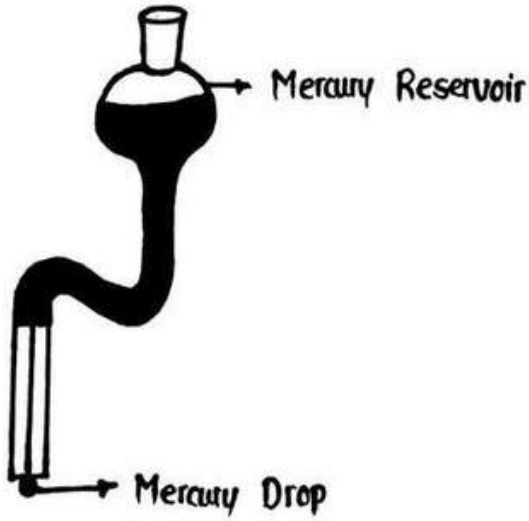
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DROPPING MERCURY ELECTRODE

- A Dropping Mercury Electrode is an electrode commonly used in polarography.
- It consist of a thin capillary tube from which mercury drops continuously and slowly fall into the solution under gravity.
- Each drop provides a fresh, clean and smooth surface for electrochemical reactions.

WORKING:

- The DME functions by continuously forming mercury drops at the tip of a fine capillary tube under the influence of gravity.
- These drops fall at regular intervals, typically every few seconds. providing a clean, renewable electrode surface.
- A linearly increasing potential is applied b/w the DME and reference electrode.
- As voltage increase, analyte undergoes oxidation or reduction (electron transfer)
- This reaction generates an electric current, which is then recorded for identifying the concentration of analyte



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